

Shubhit Sardana

Curriculum Vitae

Department of Physics
Indian Institute of Science Education and Research
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Education

Dec 2021 – June 2026 **BS-MS Dual Degree (Bachelor & Master of Science)**, Indian Institute of Science Education and Research (IISER) - Bhopal, Madhya Pradesh - 462066, India.

MS coursework average (2-semesters/1-year): 9.27/10
BS coursework average (8-semesters/4-years): 8.26/10
Cumulative Performance Index (CPI): 8.48/10 (overall)

Publications

Total Citations: 43, Published: 4, Pre-print: 1, **INSPIRE-HEP**

In Order of Relevance

- Mar 2026 **[1] Sardana, S.**, Goncharov, B., & Tremblay, J. C. (2026). The Targeted Standard Siren Cosmology with Pulsar Timing Arrays. <https://arxiv.org/abs/2603.12168>, arXiv:2603.12168 [astro-ph.CO]. Submitted: **Nature Astronomy** – 1st Author
- Oct 2025 **[2]** Goncharov, B., **Sardana, S.**, Sesana, A. et al. Reading signatures of supermassive binary black holes in pulsar timing array observations. Nat Commun 16, 9692 (2025). <https://doi.org/10.1038/s41467-025-65450-3>, Published: **Nature Communications** – 2nd Author
- Feb 2025 **[3]** Goncharov, B., & **Sardana, S.** (2025). Ensemble noise properties of the European Pulsar Timing Array. Monthly Notices of the Royal Astronomical Society, 537(4), 3470–3479. <https://doi.org/10.1093/mnras/staf190>, Published: **Monthly Notices of the Royal Astronomical Society (MNRAS)** – 2nd Author
- Dec 2025 **[4]** Nobleson, K., Dwivedi, C., Desai, S., et al. (2025), The Indian Pulsar Timing Array Data Release 2: II. Customised Single-Pulsar Noise Analysis and Noise Budget, <https://doi.org/10.1016/j.jheap.2026.100594>, Published: **Journal of High Energy Astrophysics (JHEAP)** – InPTA Collaboration
- Aug 2025 **[5]** Ojha, A., **Sardana, S.**, & Ghosh, A. (2025). Statistical mechanics from quantum envariance and exchange symmetry. <https://doi.org/10.1103/dh9r-4ybyq>, Published: **Physical Review A** – 2nd Author

Research Experiences

- June 2025 – May 2026 **Master's Thesis**, Indian Institute of Science Education and Research, Bhopal, India.
Thesis Supervisor: Dr. Mayuresh Surnis, IISER Bhopal, India
Co-Supervisor: M. A. Krishnakumar, NCRA, TIFR, India
Topic: Using IPS to understand PTA Noise Budget
- Used interplanetary scintillation (IPS) techniques to quantify heliospheric electron density variations and associated dispersion measure (DM) fluctuations in signal.
 - Developed tomographic framework to understand the heliospheric plasma density variation over a time period of eight months.
 - Utilised eight months of Ooty Radio Telescope (ORT) archival observations to construct heliospheric plasma density maps (3d tomography).

- August 2025 – **Research Intern, Max Planck Institute for Gravitational Physics, Hannover, Germany.**
 March 2026 **Supervisor:** Dr. Boris Goncharov
Topic: The Targeted Standard Siren Cosmology with Pulsar Timing Arrays
- Developed parallelized MCMC pipelines for PTA-based CGW analysis, incorporating custom noise models and simultaneous multi-source searches.
 - Implemented GPU-accelerated Bayesian inference using JAX and NumPyro (HMC) for high-dimensional cosmological parameter estimation.
 - Applied Normalizing Flows for fast posterior estimation of SMBHB parameters to improve inference efficiency.
 - Combined CGW-inferred luminosity distances with galaxy catalog redshifts to constrain cosmological parameters, achieving $\sim 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ precision on H_0 .
- Feb 2025 – **Research Collaboration, Indian Pulsar Timing Array (InPTA).**
 June 2025 **Noise Analysis Group, InPTA**
Topic: Incorporating Solar Wind Gaussian Process in Pulsar Noise Modeling
- Incorporated solar wind effects into pulsar noise models for low solar elongation pulsars in the InPTA DR2 dataset.
 - Developed data-analysis pipelines using a Gaussian-process framework to model solar wind-induced timing variations and validated the approach across multiple pulsars.
 - Regularly presented progress and contributed to methodological refinements through collaborative discussions in weekly group meetings.
- May 2024 – **Research Intern, Max Planck Institute for Gravitational Physics, Hannover, Germany.**
 Aug 2024 **Supervisor:** Dr. Boris Goncharov (Senior Scientist)
Topic: Ensemble noise properties of the European Pulsar Timing Array
- Implemented a novel prior-reweighting methodology developed by Dr. Boris Goncharov to marginalize over the uncertainties in pulsar noise priors using the EPTA DR2 dataset.
 - Performed Hierarchical Bayesian inference to infer GWB parameters, enabling a more robust and statistically consistent characterization of PTA noise properties.
 - Demonstrated that incorporating updated prior distributions increases the bayesian evidence for the gravitational-wave background (GWB) by improving consistency between inferred GWB parameters and theoretical expectations while mitigating systematic measurement biases.
 - This work resulted in two peer-reviewed research publications: “*Ensemble Noise Properties of the European Pulsar Timing Array*” (MNRAS) and “*Reading Signatures of Supermassive Binary Black Holes in Pulsar Timing Array Observations*” (Nature Communications).
- Dec 2023 – **Research Intern, Ludwig Maximilian University of Munich (LMU Munich), Munich, Germany.**
 Apr 2024 **Supervisor:** Dr. Karan Molaverdikhani
Topic: Analysis of EVLA Archival Data for Exoplanet Host Star Radio Emissions
- Analyzed the TRAPPIST-1 system to search for potential star–planet interaction signatures in radio wavelengths using archival EVLA observations.
 - Processed, calibrated, and imaged extensive radio datasets using CASA and standard VLA data-reduction pipelines, followed by detailed analysis from the resulting images.
 - Found no definitive evidence for radio emission associated with star–planet interactions, likely due to sensitivity limitations of current radio telescopes.
 - Identified next-generation facilities such as the ngVLA and SKA-Mid as promising instruments capable of achieving the sensitivity required for future detections of such interactions.
- May 2023 – **Research Intern, Indian Institute of Science Education and Research, Bhopal, India.**
 Dec 2023 **Supervisor:** Dr. Mayuresh Surnis
Topic: Pulsar Search and Analysis Pipeline Development for GMRT Data
- Built and optimized a PRESTO-based pulsar search pipeline for GMRT data, including filterbank generation, RFI mitigation (RFI Clean), and de-dispersion.
 - Improved computational performance by replacing single-frequency shifting with a sub-band shifting scheme, reducing I/O overhead while preserving accuracy.
 - Performed accelerated pulsar searches using `accelsearch` and validated pulsar candidates.
 - Acquired hands-on experience in pulsar signal processing, pulse profile analysis, and end-to-end detection workflows.

Conference & Teleconference Presentations

- May 20, 2026 **European Pulsar Timing Array (EPTA)-wide Teleconference.**
Topic: (Research Paper) The Targeted Standard Siren Cosmology with Pulsar Timing Arrays. Presented results from my first author paper focusing on unlocking the potential of PTAs as a cosmological probe to constrain the cosmological constants.
- Mar 5, 2025 **Indian Pulsar timing Array (InPTA)-wide Teleconference.**
Topic: (Research Paper) Exploring the time variability of the solar wind using LOFAR pulsar data. Presented on the importance of solar wind in PTA. Mentioned the need for integrating these effects and presented the results from the script built by me to analyze the process.
- Oct 17, 2024 **International Pulsar Timing Array (Gravitational-wave Analysis) Teleconference.**
Topic: (Two Research Papers) Ensemble noise properties of pulsars in EPTA and implications. GWA is one of the several working groups in the IPTA. The presentation focused on the key findings from the work done under the supervision of Dr. Boris Goncharov. This work has also been published in the form of two research papers.
- July 29, 2024 **European Pulsar Timing Array (EPTA)-wide Teleconference.**
Topic: (Research Paper) Ensemble noise properties of pulsars in EPTA. Presented in the EPTA-wide Telecon, a bi-weekly EPTA meeting. The presentation focused on the Ensemble noise properties of Pulsars, a research project that I did under the supervision of Dr. Boris Goncharov.

Technical Skills

- Programming Python (NumPy, Pandas, SciPy, Matplotlib) – Proficient, C/C++ – Intermediate
- GPU-based PyTorch, JAX, Normalizing Flows (Variational Inference), Discovery-Nanograv (GPU) programming
- Statistical Bayesian Inference, Markov Chain Monte Carlo (MCMC), Hamiltonian Monte Carlo (HMC), Tools Basic Frequentist Inference
- Astronomy Astropy, CASA, WSClean (Interferometric Imaging), PRESTO, Tempo2 Softwares
- Other Tools Linux – Proficient, High-Performance Computing (HPC) – Proficient, Bash/Shell Scripting – Intermediate, Mathematica, HTML5, CSS

Advanced Coursework

- Astrophysics & Relativity Astronomy and Astrophysics, Radio Interferometry (NRAO Curriculum – self-studied), Cosmology, General Theory of Relativity, Special Relativity.
- High-Energy & Particle Electrodynamics, Nuclear and Particle Physics, Atomic and Molecular Physics.
- Computation & Mechanics Advanced Quantum Mechanics, Statistical Mechanics, Non-Linear Dynamics and Chaos, Classical Mechanics, Numerical Methods and Programming, Mathematical Methods I & II.

Workshops & Certifications

- o Attended the two-day *Winter Conference on Condensed Matter Physics, Astrophysics & High Energy Physics (Seventh Edition, 2024)*, conducted online and organized by The University of Tennessee, Knoxville, USA.
- o Participated in the *Sagan Summer Workshop 2023*, a five-day online workshop hosted by the NASA Exoplanet Science Institute, Caltech, Pasadena, CA.
- o Attended a two-week online workshop, *Overview of Space Science (START Program)*, organized by the Indian Space Research Organisation (ISRO).
- o Completed the Coursera course *The Evolving Universe*, taught by Dr. S. George Djorgovski (California Institute of Technology), focusing on the evolution of the Universe from early epochs to large-scale structure formation.